

File Systems and Storage -- Guided Notes ANSWER KEY

Unit tp-5.2 | CompTIA Network+ N10-009 Obj. FC0-U71 5.2 | N10-008 FC0-U71 5.2

For instructor use / distribute after students complete the notes.

Question 1

A file system organizes how data is _____, named, and accessed on a storage device. Without a file system, a drive is just a _____ series of bytes. Common Windows file systems include _____ (legacy FAT) and _____ (modern, supports large files and permissions). Linux commonly uses _____.

Answer: Stored; raw (undifferentiated); FAT32 (exFAT); NTFS; ext4.

Question 2

FAT32 has a maximum single file size of _____ GB. NTFS supports files up to _____ TB theoretically. exFAT was designed for _____ drives and supports large files without the 4 GB FAT32 limit. NTFS supports features FAT32 lacks: _____ (file permissions), _____ (access logs), and file-level encryption.

Answer: 4 GB; 16 TB+; flash (USB/SD); permissions; auditing.

Question 3

A _____ is the directory hierarchy root from which all files are organized. On Windows, drive letters like _____ are the root for each volume. On Linux and macOS the entire filesystem has a single root called _____. Linux uses _____ as the path separator; Windows uses _____.

Answer: Root; C:\ (or D:\); / (forward slash); /; \ (backslash).

Question 4

A file _____ (like .txt, .docx, .exe) tells the OS what type of data the file contains and which program to open it with. Extensions are not embedded in the data -- they are just part of the _____ and can be changed. On Windows, the _____ of a file is stored in NTFS metadata, not in the extension.

Answer: Extension; filename; file type (MIME type).

Question 5

Compression reduces _____ size by encoding repeated patterns more efficiently. _____ compression (ZIP, gzip) reduces size without losing data and can be restored to the original. _____ compression (JPEG, MP3) permanently discards data to achieve greater size reduction. Lossy compression is acceptable for _____ where small quality loss is imperceptible.

Answer: File; lossless; lossy; multimedia (images/audio).

Question 6

Encryption converts file data into _____ text that cannot be read without the _____ key. BitLocker (Windows) and FileVault (macOS) are full-disk _____ solutions that encrypt the entire drive. If the drive is lost or stolen, the data cannot be read without the _____ key or recovery code.

Answer: Cipher; decryption; encryption; encryption (BitLocker).

Question 7

A _____ is a file or folder shortcut (symbolic link on Linux/macOS, or shortcut .lnk on Windows) that points to another location without copying the data. Deleting the shortcut does not delete the _____ file. On Linux, a _____ link is a direct pointer (inode reference) to the same data, so deleting one does not delete the other as long as another link exists.

Answer: Link (shortcut); original; hard.

Question 8 -- Real World Application

REAL WORLD: A student copies a 5 GB video file from their laptop to a FAT32 USB drive and receives an error saying the file is too large even though the drive has 16 GB of free space. Explain why this error occurs and describe two ways to solve it.

Answer: FAT32 has a 4 GB per single file size limitation, regardless of how much total free space exists on the drive. The 5 GB file exceeds this limit, so the copy fails. This is an architecture limitation of the FAT32 file system -- it uses a 32-bit field to store file size, which caps at $2^{32} - 1$ bytes (approximately 4 GB). Solution 1 -- Reformat the USB drive to exFAT: exFAT was designed for flash drives and supports files much larger than 4 GB with no practical limit. The drive should be reformatted to exFAT (Windows: right-click drive > Format > exFAT). Note: reformatting erases all data. Solution 2 -- Split the video into parts smaller than 4 GB using a video splitter tool, copy both parts to the FAT32 drive, then reassemble on the destination computer. This is a workaround that keeps the FAT32 format if the destination device requires it.